



Matrix Theory

13 Orthogonal Complement

Dr. Konda Reddy Mopuri
Dept. of AI, IIT Hyderabad
July-Nov 2026

Fundamental theorem of Orthogonality



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

The row space is orthogonal to null space (in $\mathbb{R}^n$). The column space is orthogonal to the left null space (in $\mathbb{R}^m$).

Example

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$



Observations



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

- $N(A) \perp C(A^T)$

Observations



- $N(A) \perp C(A^T)$
- In fact, $N(A)$ contains every vector perpendicular to $C(A^T)$



- In general, the dimension of $N(A)$ and $C(A^T)$ add to the dimension of the ambient space



- In general, the dimension of $N(A)$ and $C(A^T)$ add to the dimension of the ambient space
- Similarly for $N(A^T)$ and $C(A)$

Orthogonal Complement



Given a subspace $\mathbf{V}$ of $\mathbb{R}^n$, the space of all vectors orthogonal to $\mathbf{V}$ is called the Orthogonal Complement of $\mathbf{V}$. It is denoted by $\mathbf{V}^\perp = \mathbf{V} \text{ perp.}$

Orthogonal Complement



Given a subspace $\mathbf{V}$ of $\mathbb{R}^n$, the space of all vectors orthogonal to $\mathbf{V}$ is called the Orthogonal Complement of $\mathbf{V}$. It is denoted by $\mathbf{V}^\perp = \mathbf{V}^\perp$ perp.

- $N(A) = (C(A^T))^\perp$

Orthogonal Complement



Given a subspace $\mathbf{V}$ of $\mathbb{R}^n$, the space of all vectors orthogonal to $\mathbf{V}$ is called the Orthogonal Complement of $\mathbf{V}$. It is denoted by $\mathbf{V}^\perp = \mathbf{V}^\perp$ perp.

- $N(A) = (C(A^T))^\perp$
- Also, the row space contains every vector orthogonal to null space,
 $C(A^T) = (N(A))^\perp$

Orthogonal Complement

Given a subspace $\mathbf{V}$ of $\mathbb{R}^n$, the space of all vectors orthogonal to $\mathbf{V}$ is called the Orthogonal Complement of $\mathbf{V}$. It is denoted by $\mathbf{V}^\perp = \mathbf{V}^\text{perp}$.

- $N(A) = (C(A^T))^\perp$
- Also, the row space contains every vector orthogonal to null space,
 $C(A^T) = (N(A))^\perp$
- Similarly for the $N(A^T)$ and $C(A)$ pair.

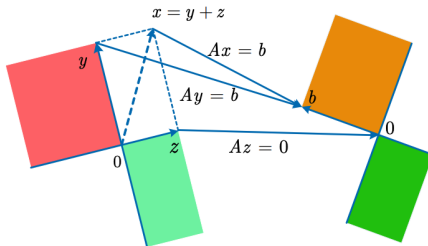
Fundamental Theorem of LA, Part II



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

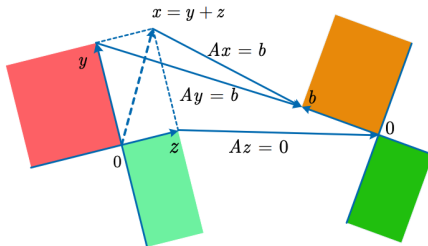
The nullspace is the orthogonal complement of the row space in $\mathbb{R}^n$.
The left nullspace is the orthogonal complement of the column space in $\mathbb{R}^m$.

Some notes



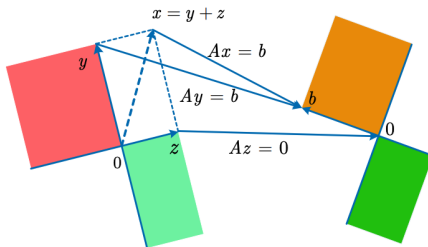
- The matrix can operate on any input in the ambient space

Some notes



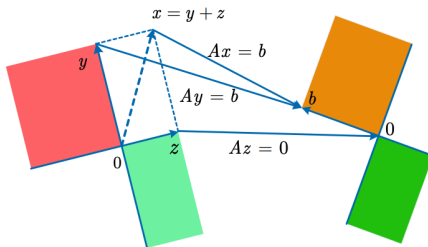
- The matrix can operate on any input in the ambient space
- Null space is carried to zero

Some notes



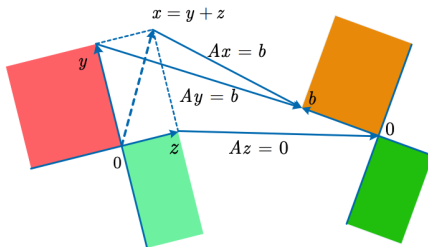
- The matrix can operate on any input in the ambient space
- Null space is carried to zero
- Every vector Ax is in the column space

Some notes



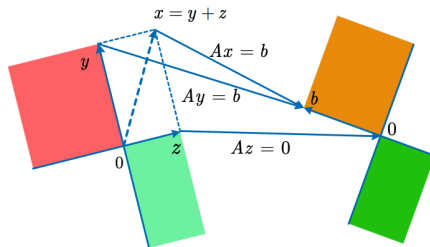
- The matrix can operate on any input in the ambient space
- Null space is carried to zero
- Every vector Ax is in the column space
- Nothing goes to $N(A^T)$

Some notes



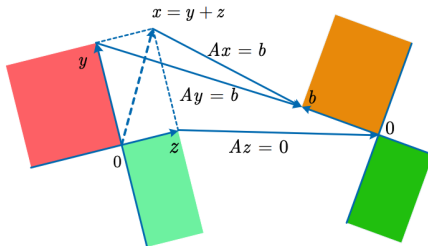
- There can be vectors outside $C(A^T)$ and $N(A)$ (i.e., their union need not fill all of $\mathbb{R}^n$).

Some notes



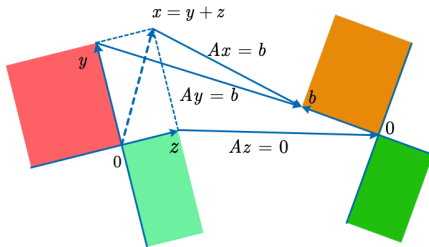
- There can be vectors outside $C(A^T)$ and $N(A)$ (i.e., their union need not fill all of $\mathbb{R}^n$).
- However, their sum does

Some notes



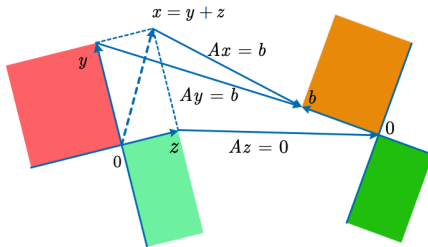
- Every x splits uniquely as $x = x_r + x_n$, with x_r in the row space and x_n in the nullspace

Some notes



- Every x splits uniquely as $x = x_r + x_n$, with x_r in the row space and x_n in the nullspace
- x_n goes to zero: $Ax_n = 0$

Some notes



- Every x splits uniquely as $x = x_r + x_n$, with x_r in the row space and x_n in the nullspace
- x_n goes to zero: $Ax_n = 0$
- x_r goes to $C(A)$: $Ax_r = Ax$

Some notes



From the $C(A^T)$ to $C(A)$, matrix A is invertible. Every vector b in the column space comes from exactly one vector x_r in the row space.

Next



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

- Projections

Rough Work



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad